

9. (a) (i) The average bond enthalpy of a C—C bond is quoted as 348 kJ mol^{-1} .

Explain what is meant by *bond enthalpy*.

[2]

- (ii) Ethyne, C_2H_2 , contains a $\text{C}\equiv\text{C}$. It reacts with hydrogen in a similar way to ethene.



Some average bond enthalpies are given in the table.

Bond	Average bond enthalpy / kJ mol^{-1}
$\text{C}\equiv\text{C}$	839
$\text{C}-\text{C}$	348
$\text{C}-\text{H}$	413
$\text{H}-\text{H}$	436

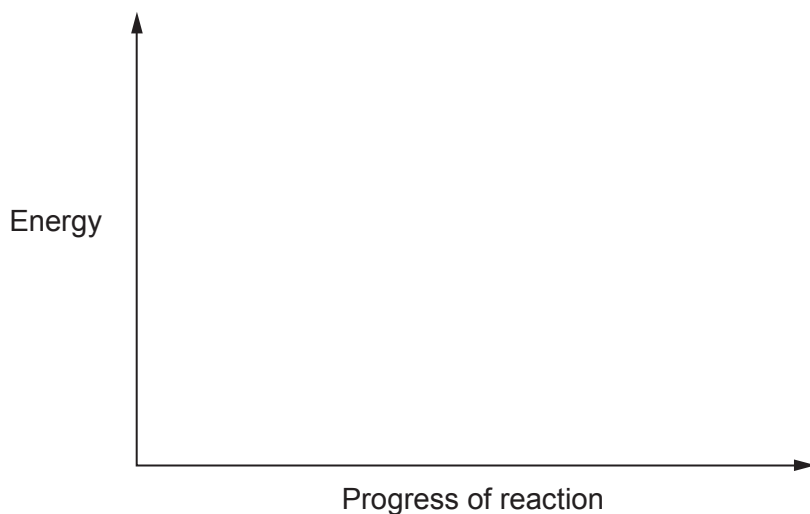
Use the data to calculate the enthalpy change, ΔH , for the reaction of ethyne and hydrogen.

[3]

$\Delta H = \dots\dots\dots \text{ kJ mol}^{-1}$



- (iii) Use your answer to part (ii) to sketch an energy profile diagram for this reaction on the axes below. Label ΔH and the activation energy, E_a , on your diagram. [2]

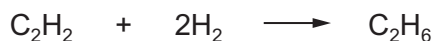


- (b) Enthalpy changes of reaction are often found indirectly. The enthalpy change for the reaction of ethyne with hydrogen, as shown in part (a), can be determined by using enthalpy changes of combustion.

The table gives some enthalpy changes of combustion, $\Delta_c H^\theta$.

Substance	Enthalpy change of combustion, $\Delta_c H^\theta$ / kJ mol^{-1}
hydrogen, H_2	-286
ethyne, C_2H_2	-1300
ethane, C_2H_6	-1600

Use these enthalpy changes to calculate the enthalpy change, ΔH , for the reaction of ethyne and hydrogen. [3]



$\Delta H = \dots\dots\dots \text{kJ mol}^{-1}$



- (c) The theoretical values that you have calculated in parts (a)(ii) and (b) are both the enthalpy change for the reaction between ethyne and hydrogen.

Suggest a reason why these values are not the same. [1]

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- (d) Suggest the type of reaction that occurs between ethyne and hydrogen. [1]

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